



Antimicrobial Resistance of *Shigella flexneri* in Pakistani Pediatric Population Reveals an Increased Trend of Third-Generation Cephalosporin Resistance

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Abstract

The rapid emergence of resistance to third-generation cephalosporins in *Shigella flexneri* is crucial in pediatric shigellosis management. Limited studies have been conducted on molecular pattern of antibiotic resistance of *S. flexneri* in diarrhea endemic areas of Pakistan. The aim of the study was to analyze the antimicrobial resistance of *S. flexneri* isolated from pediatric diarrheal patients in Peshawar, Pakistan. A total of 199 *S. flexneri* isolates (clinical, $n = 155$ and non-clinical, $n = 44$) were investigated for drug resistance and mutational analysis of selected drug resistance genes. All isolates were found to be highly resistant to amoxicillin/clavulanic acid (88%), followed by trimethoprim–sulfamethoxazole (77%), chloramphenicol (43%), and quinolones (41.6%). About 34.5% *S. flexneri* isolates were found to be resistant to third-generation cephalosporin. None of the isolates was resistant to imipenem, piperacillin-tazobactam, and amikacin. Interestingly high frequency of third-generation cephalosporin resistance was observed in *S. flexneri* isolated from non-clinical samples (49%) when compared to clinical samples (30.5%). Furthermore, the most prevalent phenotypic-resistant patterns among third-generation cephalosporin-resistant isolates were AMC,CAZ,CPD,CFM,CRO,SXT (13%) followed by OFX,AMC,CAZ,CPD,CFM,CRO,SXT,NA,CIP (10%). The most frequently detected resistance genes were trimethoprim–sulfamethoxazole ($sul2 = 84\%$), beta-lactamase genes ($bla_{OXA} = 87\%$), quinolones ($qnrS = 77\%$), and chloramphenicol ($cat = 64\%$). No mutation was detected in any drug-resistant genes. We are reporting for the first time the sequence of the bla_{TEM} gene in *S. flexneri*. Furthermore, high third-generation cephalosporin resistance was observed in the patients who practiced self-medication as compared to those who took medication according to physician prescription. This study shows the high emergence of third-generation cephalosporin-resistant *S. flexneri* isolates, which is a potential threat to the community in the country. This finding will be helpful to develop a suitable antibiotic prescription regime to treat shigellosis.

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Introduction

Shigellosis is one of the major causes of illness and death in pediatrics throughout the world [1]. Annually approximately 89.9 million individuals have been affected from shigellosis and 1.1 million deaths occur globally [2]. Among pediatrics in developing countries the estimated annual death rate due to shigellosis was between 28,000 and 48,000 [3]. Antibiotic treatment is commonly suggested for shigellosis as it reduces the infection duration and severity and prevent complications [4]. But *S. flexneri* has acquired a rapid increase in resistance against various antibiotics worldwide and this has emerged as a therapeutic challenge [4, 5]. In the last few years, multidrug resistance (MDR) in *S. flexneri* has been increased. MDR phenotype attains by various mechanisms but the most common one for the increase of antibiotics

resistance is the horizontal transmission of genetic factors [6]. The ampicillin, chloramphenicol, trimethoprim/sulfamethoxazole, and tetracycline are no longer suggested as first-line drugs for shigellosis treatment due to the rapid development of antibiotics resistance. Moreover cephalosporin are used as an alternative of quinolones to treat shigellosis in adults and children but the emergence of cephalosporin resistance among *S. flexneri* isolates becoming a challenge to physician worldwide especially in developing countries [7]. First *S. flexneri*-resistant isolate against third-generation cephalosporin was reported in Paris in 1995 [8]. Recently sporadic third-generation cephalosporin-resistant isolates of *S. flexneri* were reported from various parts of the world including Pakistan that limiting the therapeutic options [9–12]. A gradually increase in third-generation cephalosporin resistance was observed in different countries. In Pakistan, 2.4% cephalosporin-resistant isolates of *Shigella* have been reported from 1996 to 2007 while 20% and 19.5% isolates were resistant to at least one of the third-generation cephalosporin in 2012 and 2014, respectively [13–15]. Similarly resistance to the third-generation cephalosporin have been reported from South-East Asia (2.0–5.2%), India (16.8%), China (18%), and Iran (18.75%) [4, 16, 17]. MDR *S. flexneri* has become a challenge in the management of shigellosis in developing country particularly in Pakistan. There is no comprehensive data available on the recent trends of MDR and third-generation cephalosporin resistance among *S. flexneri* in Pakistan. In our previous study, the pattern of antimicrobial resistance among different serotypes of *S. flexneri* were investigated [18]. Now in the current study, we analyzed the antimicrobial resistance of *S. flexneri* isolated from clinical settings and non-clinical source (Drinking water and raw food) in Peshawar, Pakistan. The findings of the present study will be helpful to physicians in formulating effective antibiotics prescription strategies to control shigellosis cases in Peshawar, Pakistan.

Methodology

Bacterial Isolates

The present study includes 199 *S. flexneri* strains including 155 clinical strains isolated from stool samples collected from different diarrheal patients with mean age of 9.12 before any prescribed antibiotic treatment visiting at various tertiary care hospitals of Peshawar and 44 non-clinical strains isolated from different various non-clinical sources (household drinking water, retail raw milk, and fruits/vegetables) from different towns of Peshawar. All these *S. flexneri* strains were previously characterized by 16S rRNA sequencing and molecular serotyping [18–20]. Both

clinical and non-clinical samples were further processed for analysis of antibiotic susceptibility testing.

Pre-informed consent was taken from patients or guardians for the collection of samples and associated data. Respondents were questioned whether they had taken an antibiotic without being prescribed it by a physician's (self-medication) before 4 or more than 4 days of study enrollment; frequency of antibiotic had taken without prescription. The present study was approved by the institutional Bioethical committee.

Antibiotic Susceptibility Testing of *S. flexneri*

A total of 199 characterized *S. flexneri* strains were processed for analysis of antibiotic susceptibility assay. In vitro activities, the antimicrobial susceptibility assay was performed by Kirby–Bauer disk diffusion method following the recommended instructions by Clinical and Laboratory Standards Institute [21]. The standard antibiotic disks (OXOID) used in the study were: trimethoprim–sulfamethoxazole (SXT, 30 µg), chloramphenicol (C, 30 µg), piperacillin-tazobactam (TZP, 110 µg), amoxicillin/clavulanic acid (AMC, 30 µg), imipenem (IPM, 10 µg), ciprofloxacin (CIP, 5 µg), ofloxacin (OFX, 5 µg), ceftazidime (CAZ, 30 µg), cefpodoxime (CPD, 10 µg), cefixime (CFM, 5 µg), ceftriaxone (CRO, 30 µg), nalidixic acid (NA, 30 µg), and amikacin (AK, 30 µg). In the present study, MDR is defined as resistance to three or more unrelated antibiotic classes.

Screening of Antibiotic Resistance Genes

Antibiotic resistance genes comprising plasmid-mediated quinolone resistance (*qnrA*, *qnrB*, *qnrS*, *aac*), beta-lactam genes (*blaOXA*, *blaTEM*, and *blaSHV*), chloramphenicol (*cat*) and sulfonamide (*sul1* and *sul2*) were screened using PCR assay as described earlier [22]. All PCR amplifications were done using the 2×*phire* green hot start II PCR master mix reagent (ThermoFisher, CA, USA) according to vendor instructions. The primers used in this study are described previously [22–27]. In this study PCR grade water (ThermoFisher scientific, St. Leon-Rot, Germany) was used as a negative control. Thermal cycling was carried out by an initial denaturation step at 98 °C for 90 s, followed by 32 cycles of denaturation at 98 °C for 10 s, annealing at a different temperature as described in supplementary Table S1 for 10 s, and elongation at 72 °C for 45 s which was followed by a final elongation step at 72 °C for 55 s in a thermocycler (Senso, Germany). A portion (5 µl) of the amplified product was mixed with loading dye, subjected to electrophoresis in 1.5% agarose gel, and visualized in UV transilluminator.

Sequencing of Amplified Drug-Resistant Gene and Bioinformatics Analysis

The amplified product was then purified using GenEluteTM PCR Clean-up Kit (NA1020, Sigma Aldrich, USA). Purified PCR product was sequenced by Macrogen Inc. (Amsterdam, The Netherlands) and was then analyzed by clone manager suite 7 and National Centre of Biotechnology Information (NCBI) online tools using the BLAST (blastx).

Statistical Analysis

Pearson's χ^2 test was performed to evaluate the association of (i) drug-resistant *S. flexneri* with particular emphasis on third-generation cephalosporin-resistant isolates among clinical and non-clinical samples. (ii) Multidrug resistance patterns among the *S. flexneri* strains isolated from clinical and non-clinical sources. (iii) Drug resistance genes among the clinical and non-clinical isolates of *S. flexneri*. *P* values less than 0.05 were considered to be statistically significant.

Results

Antibiotic Susceptibility Pattern

A disk diffusion assay was performed to examine the antibiotic susceptibility of 199 *S. flexneri* isolates to an array of 13 antibiotics. All these isolates were observed to be highly resistant to designated drugs. Of these 88% of the isolates were resistant to amoxicillin/clavulanic acid followed by trimethoprim-sulfamethoxazole (77%). About 41.6% isolates were resistant to quinolones (CIP, OFX, and NA) and 43% isolates were resistant to chloramphenicol (Table 1). The frequency of isolates resistant to a third-generation antimicrobial cephalosporin (CRO, CAZ, CPD, and CFM) was 34.5%. All of the isolates were found to be sensitive to imipenem, piperacillin-tazobactam, and amikacin (Table 1).

We also observed 90%, 81%, and 46% of clinical isolates were resistant to amoxicillin/clavulanic acid, trimethoprim-sulfamethoxazole, and chloramphenicol, respectively, but 82%, 61%, and 32% of non-clinical isolates were resistant to amoxicillin/clavulanic acid,

Table 1 Prevalence of phenotypic antibiotic resistance profiles among *S. flexneri* isolated from clinical (pediatric diarrheal stool) and non-clinical (retail raw food and drinking water) samples

Antibiotics	Drug susceptibility	No. of isolates, <i>n</i> (%)	Clinical isolates, <i>n</i> (%)	Non-clinical isolates		
				Drinking water isolates, <i>n</i> (%)	Retail raw milk isolates, <i>n</i> (%)	Total isolates, <i>n</i> (%)
AMC (30 µg)	Susceptible	23 (12)	15 (10)	3 (14)	1 (8)	8(18)
	Resistant	176 (88)	140 (90)	19 (86)	11 (92)	36(82)
CAZ (30 µg)	Susceptible	129 (65)	110 (71)	14 (64)	5 (42)	19(43)
	Resistant	70 (35)	45 (29)	8 (36)	7 (58)	25(57)
CPD (10 µg)	Susceptible	125 (63)	106 (68)	14 (64)	5 (42)	19(43)
	Resistant	74 (37)	49 (32)	8 (36)	7 (58)	25(57)
CFM (5 µg)	Susceptible	129 (65)	107 (69)	13 (59)	3 (25)	22(50)
	Resistant	70 (35)	48 (31)	9 (41)	9 (75)	22(50)
CRO (30 µg)	Susceptible	138 (69)	108 (70)	17 (77)	6 (50)	30(68)
	Resistant	61 (31)	47 (30)	5 (23)	6 (50)	14(32)
SXT (30 µg)	Susceptible	46 (23)	29 (19)	10 (45)	4 (33)	17(39)
	Resistant	153 (77)	126 (81)	12 (55)	8 (67)	27(61)
OFX (5 µg)	Susceptible	118 (59)	92 (59)	12 (55)	9 (75)	26(59)
	Resistant	81 (41)	63 (41)	10 (45)	3 (25)	18(41)
CIP (5 µg)	Susceptible	115 (58)	92 (59)	11 (50)	6 (50)	23 (52)
	Resistant	84 (42)	63 (41)	11 (55)	6 (50)	21 (48)
NA (30 µg)	Susceptible	115 (58)	89 (57)	12 (55)	9 (75)	26 (59)
	Resistant	84 (42)	66 (43)	10 (45)	3 (25)	18 (41)
C (30 µg)	Susceptible	113 (57)	83 (54)	15 (68)	9 (75)	30 (68)
	Resistant	86 (43)	72 (46)	7 (32)	3 (25)	14 (32)

AMC Amoxicillin/clavulanic acid, CAZ Ceftazidime, CPD Cefpodoxime, CFM Cefixime, CRO Ceftriaxone, SXT Trimethoprim-sulfamethoxazole, OFX Ofloxacin, CIP Ciprofloxacin, NA Nalidixic acid, C Chloramphenicol

trimethoprim–sulfamethoxazole, and chloramphenicol, respectively. We noticed a high number of non-clinical isolates (49%) of *S. flexneri* were resistant to a third-generation cephalosporin (CRO, CAZ, CPD, and CFM) as compared to clinical isolates (30.5%). Similarly, we also observed 41.6% of clinical isolates were resistant to quinolones (CIP, OFX, and NA) as compared to non-clinical isolates (43.3%) (Table 1).

Furthermore, the multidrug resistance (MDR) was studied yielding a diverse phenotypic antibiotic resistance pattern which was observed in 95% of the *S. flexneri* isolates obtained from clinical and non-clinical samples. The most dominant MDR patterns observed in clinical isolates was AMC,SXT,C (34%, $P=0.01$) and in non-clinical isolates was AMC,OFX,NA,CIP (25%, $P=0.01$) as shown in Table 2. The current study also showed that the pattern only found in clinical isolates were OFX,AMC,NA,CIP,C that had a significant difference ($P=0.04$) (Table 2).

We observed no significant association among the phenotypic-resistant patterns of third-generation cephalosporin-resistant isolates isolated from clinical samples and non-clinical samples as described in Table 2.

Molecular Analysis of the Antibiotic Resistance Gene

All phenotypic-resistant isolates were studied for the presence of 10 different antibiotic resistance genes using PCR assay which yielded the expected fragments belonging to specific antibiotic resistance genes. The amplified products were then sequenced to detect the mutations in the resistant genes belonging to specific fragments patterns. blaSHV, *sul1*, *qnrA*, *qnrB*, and *aac* was not detected in the tested

isolates. The commonly known resistance mechanism against third-generation cephalosporin was due to the existence of blaOXA and blaTEM genes which were identified in 174 (87%, $P<0.00001$) and 111 (56%, $P<0.0001$) isolates, respectively. No isolate was found positive for the presence of blaSHV like genes. In addition, the PMQR (plasmid-mediated quinolone-resistant) *qnrS* gene was identified in 77% of all quinolone-resistant isolates. The *sul2* gene was present in 168 (84%, $P=0.0004$) trimethoprim–sulfamethoxazole-resistant isolates. Resistance against chloramphenicol was generally associated with the existence of *cat* genes in 128 (64%, $P=0.001$) isolates as described in Table 3. The results revealed that a strong association existed between the most predominant genotypic resistance pattern blaOXA,blaTEM,*sul2*,*qnrS*,*cat* ($P=0.002$) of clinical and non-clinical samples as described in Table 4. The present study also revealed the blaOXA,*sul2*,*cat*,*qnrS* are the only genotypic resistance pattern with statistical relevance that was observed only in clinical isolates ($P=0.04$). Several

Table 3 Frequency of drug resistance genes in *S. flexneri* isolated from clinical (pediatric diarrheal stool) and non-clinical (retail raw food and drinking water) samples

Drug-resistant gene	Positive, <i>n</i> (%)	Negative, <i>n</i> (%)	<i>P</i> value
blaTEM	111 (56)	88 (44)	<0.0001
blaOXA	174 (87)	25 (13)	<0.0001
<i>sul2</i>	168 (84)	31 (16)	0.0004
<i>qnrS</i>	153 (77)	45 (23)	0.2222
<i>cat</i>	128 (64)	71 (36)	0.001

P value <0.05 was considered statistically significant

Table 2 Comparative multidrug resistance patterns among clinical and non-clinical *S. flexneri* isolates

Multidrug resistance pattern	Clinical isolates, <i>n</i> (%)	Non-clinical isolates, <i>n</i> (%)	<i>P</i> value
AMC,SXT,C	53(34)	7(16)	0.01
OFX,AMC,NA,CIP	16(10)	11(25)	0.01
OFX,AMC,NA,CIP,C	13(8)	0(0)	0.04
OFX,SXT,NA,CIP	7(4.5)	7(16)	0.009
AMC,CAZ,CPD,CFM,CRO,SXT	21(13.5)	5(11)	0.704
SXT	7(4.5)	2(5)	0.99
AMC,CAZ,CPD,CFM,CRO,SXT,C	5(3)	4(9)	0.09
OFX,AMC,CAZ,CPD,CFM,CRO,SXT,NA,CIP	15(10)	6(14)	0.45
AMC,CPD,CFM,CRO,SXT,NA,C	3(2)	0(0)	0.35
OFX,AMC,SXT,NA,CIP	1(0.5)	0(0)	0.59
OFX,AMC,CAZ,CPD,CIP	3(2)	0(0)	0.35
AMC,CPD,CFM,SXT,NA,C	3(2)	0(0)	0.35
AMC,CAZ,CPD,CFM,SXT	8(5)	2(5)	0.86

P value <0.05 was considered statistically significant

AMC Amoxicillin/clavulanic acid, CAZ Ceftazidime, CPD Cefpodoxime, CFM Cefixime, CRO Ceftriaxone, SXT Trimethoprim-sulfamethoxazole, OFX Ofloxacin, CIP Ciprofloxacin, NA Nalidixic acid, C Chloramphenicol

Table 4 Genotypic multidrug resistance patterns among *S. flexneri* isolated from clinical and non-clinical samples

Genotypic resistance pattern	Clinical isolates, n (%)	Non-clinical isolates, n (%)	P value
blaOXA,blaTEM,sul2,qnrS, cat	62 (40)	9 (20)	0.002
blaOXA,qnrS,cat	25 (17)	9 (20)	0.58
blaOXA,sul2,cat,qnrS	7 (4)	0	0.043
sul2,qnrS,	19 (12)	6 (14)	0.67
blaOXA,blaTEM,sul2	24 (15)	6 (14)	0.84
blaOXA,blaTEM,sul2,cat	5 (3)	3 (7)	0.194
blaOXA,sul2,qnrS	6 (4)	8 (18)	0.001
blaOXA,blaTEM,qnrS,cat	1 (1)	0	0.31
blaOXA,cat,sul2	6 (4)	3 (7)	0.34

P value < 0.05 was considered statistically significant

other genotypic resistance patterns were also observed in *S. flexneri* isolated from different samples as shown in Table 4.

The nucleotide sequences of various resistance genes were analyzed using blastx to identify the similarity to previously verified blaTEM, blaOXA, cat, sul2, and qnrS genes as shown in Table 5. The sequences of blaOXA showed a close similarity to previously described OXA-1 which was also characterized earlier in *S. flexneri* isolates. The sequences of blaTEM showed a close similarity to previously sequenced TEM-1 type beta-lactamase gene that was characterized in *Shigella sonnei*, *Acinetobacter baumannii*, *Klebsiella pneumoniae*, *Escherichia coli*, and *Salmonella enterica* but were not previously sequenced in *S. flexneri* isolates. Similarly, other tested drug-resistant genes sequences like cat, sul2 and qnrS showed close similarity with earlier described cases catA and catA1, sul2, qnrS1 and qnrS, respectively, that was characterized previously in *S. flexneri* isolates as shown in Table 5. Furthermore, no mutation was detected in any drug-resistant genes.

Antibiotic Susceptibility and Self-medication

In order to examine the antibiotic resistance is related to the fact if patients were prescribed with antibiotics or were self-medicated; the association between these two variables was studied. Patients which were self-medicated with antibiotics without prescription, before visiting the hospital showed 64.6% antibiotic resistance as compared to those who used antibiotics according to prescription (35.4%) with a significant association ($P < 0.0001$) (Table 6). In the current study, we also observed a patient which were self-medicated with third-generation cephalosporin (CRO, CAZ, CPD, and CFM) showed a high level of resistance (64.2%) as compared to those who used antibiotics according to prescription (35.82%). (Table 7).

Discussion

The current study reporting the molecular characterization of multidrug-resistant *S. flexneri* isolates and the recent trends of third-generation cephalosporin resistance among *S. flexneri* in Pakistan. We observed, a high level of third-generation cephalosporin-resistant isolates isolated from both clinical and non-clinical samples. A high level of resistance among different *S. flexneri* isolates was reported in this study against common antibiotics including quinolone, third-generation cephalosporin, amoxicillin/clavulanic acid, chloramphenicol, and sulfamethoxazole-trimethoprim drugs when compared to earlier reports from Pakistan [28, 29]. Research published in 2003 with obtained data from the northern part of Pakistan demonstrated that 10% of the *S. flexneri* isolates were sensitive to all antibiotics [28]. In 2009, in Karachi, 5.6% of all *S. flexneri* isolates were sensitive to all antibiotic drugs [30]. While in the present finding we observed no sensitive *S. flexneri* isolates to all tested antibiotic drugs. The high antibiotic-resistant is likely due to the direct outcome of uncontrolled use and easy accessibility to antibiotics in the community.

These findings strongly recommended that in case of severity of disease amoxicillin/clavulanic acid and sulfamethoxazole-trimethoprim should not be used in empirical therapy in Pakistan. Resistance toward chloramphenicol and nalidixic acid was also found in *S. flexneri*. Resistance to these two drugs has been reported earlier from different countries [31–33]. In Pakistan, these drugs are frequently used for community-acquired diarrhea due to its low cost. Resistance to these drugs is distressing since, in countries with limited resources, the acquiring drug resistance against nalidixic acid and chloramphenicol will make treatment difficult and costly for family and physicians.

In previous studies, the empiric treatment of diarrhea with cephalosporin and quinolones were recommended as a frontline antimicrobial drug in which resistant to third-generation cephalosporin was a rare finding [12, 34, 35]. However, in our finding 41.6% isolates were showed resistance to quinolones, which will severely threaten the efficiency of the therapy [36, 37].

In present finding 34.5% isolates were showed resistance to at least one of cephalosporin (CRO, CAZ, CPD, and CFM) which is consistent with the study reported in Faisalabad, Pakistan that was conducted in 2009–2010 [15]. Similarly, studies conducted in Iran and India a rapid emergence of third-generation cephalosporin were also reported [3, 38].

The observed antibiotic-resistant pattern showed that none of *S. flexneri* isolates were sensitive to all tested

Table 5 Similarity index of *OXA*, *TEM*, *qnrS*, *sul2* and *cat* in previously reported cases

Similarities with reported case						
Drug resistance gene	No. of nucleotides	Bacteria	Percentage similarity	Query Cover	Reported gene	Gene bank Accession No
<i>blaOXA</i>	885	<i>Shigella flexneri</i>	100	98	<i>OXA-1</i>	WP_094168942.1
	887	<i>Shigella flexneri</i>	100	98	<i>OXA-1</i>	WP_094168942.1
	881	<i>Shigella flexneri</i>	100	83	<i>OXA-1</i>	WP_156670365.1
	886	<i>Shigella flexneri</i>	100	88	<i>OXA-1</i>	MLA07964.1
	886	<i>Shigella flexneri</i>	100	83	<i>OXA-1</i>	MLA07964.1
	885	<i>Shigella flexneri</i>	99.63	98	<i>OXA-1</i>	WP_112354762.1
	871	<i>Shigella flexneri</i>	99.59	88	<i>OXA-1</i>	WP_094168942.1
	887	<i>Shigella flexneri</i>	99.59	83	<i>OXA-1</i>	WP_112354762.1
	888	<i>Shigella flexneri</i>	99.59	83	<i>OXA-1</i>	MMD18043.1
	890	<i>Shigella flexneri</i>	99	95	<i>OXA-1</i>	MMD18043.1
	888	<i>Shigella flexneri</i>	98.81	95	<i>OXA-1</i>	WP_156670365.1
	888	<i>Shigella flexneri</i>	99	96	<i>OXA-1</i>	MLA07964.1
	883	<i>Shigella flexneri</i>	99	88	<i>OXA-1</i>	WP_111780065.1
<i>blaTEM</i>	851	<i>Shigella sonnei</i>	100	100	<i>TEM-1</i>	AQZ26952.1
	850	<i>Acinetobacter baumannii</i>	100	99	<i>TEM-1</i>	ADQ42774.1
	854	<i>Shigella sonnei</i>	100	99	<i>TEM-1</i>	AQZ26952.1
	856	<i>Klebsiella pneumoniae</i>	100	99	<i>TEM-1</i>	AXL94269.1
	857	<i>Escherichia coli</i>	100	99	<i>TEM-1</i>	AHB36963.1
	854	<i>Escherichia coli</i>	100	99	<i>TEM-1</i>	AHB36963.1
	858	<i>Klebsiella pneumoniae</i>	100	98	<i>TEM-1</i>	ADB90404.1
	855	<i>Shigella sonnei</i>	100	97	<i>TEM-1</i>	ADY02598.1
	858	<i>Escherichia coli</i>	100	96	<i>TEM-1</i>	AHB36963.1
	856	<i>Salmonella enterica subsp. enterica</i> serovar <i>Choleraesuis</i> str. SC-B67	100	81	<i>TEM-1</i>	AAS76302.1
<i>qnrS</i>	416	<i>Shigella flexneri</i>	100	99	<i>qnrS1</i>	AJQ17368.1
	417	<i>Shigella flexneri</i>	100	99	<i>qnrS</i>	ACM67495.1
	415	<i>Shigella flexneri</i>	100	98	<i>qnrS</i>	ACM67495.1
	417	<i>Shigella flexneri</i>	100	97	<i>qnrS1</i>	ACI42280.1
	417	<i>Shigella flexneri</i>	100	94	<i>qnrS1</i>	ACI42280.1
	410	<i>Shigella flexneri</i>	100	94	<i>qnrS1</i>	AJQ17368.1
	412	<i>Shigella flexneri</i>	100	94	<i>qnrS1</i>	ACI42280.1
	417	<i>Shigella flexneri</i>	100	82	<i>qnrS1</i>	ACM67495.1
	417	<i>Shigella flexneri</i>	100	73	<i>qnrS1</i>	BBA09467.1
	413	<i>Shigella flexneri</i>	99.2	99	<i>qnrS</i>	ACM67495.1
	417	<i>Shigella flexneri</i>	99.18	98	<i>qnrS1</i>	ACM67495.1
	414	<i>Shigella flexneri</i>	98	86	<i>qnrS1</i>	AFM75687.1
<i>cat</i>	209	<i>Shigella flexneri</i>	100	96	<i>catA</i>	WP_156665709.1
	196	<i>Shigella flexneri</i>	100	95	<i>catA</i>	WP_156665709.1
	197	<i>Shigella flexneri</i>	100	95	<i>catA1</i>	WP_156665709.1
	209	<i>Shigella flexneri</i>	100	93	<i>catA1</i>	WP_156665709.1
	206	<i>Shigella flexneri</i>	100	81	<i>catA1</i>	WP_156665709.1
	197	<i>Shigella flexneri</i>	100	82	<i>catA1</i>	WP_156665709.1
	209	<i>Shigella spp.</i>	100	68	<i>catA</i>	ADY02613.1
	209	<i>Shigella spp.</i>	98.04	92	<i>catA1</i>	AEN75335.1
	209	<i>Shigella spp.</i>	99	90	<i>cat1A</i>	AEN75335.1
<i>sul2</i>	716	<i>Shigella flexneri</i>	100	99	<i>sul2</i>	EAA2119355.1
	717	<i>Shigella sonnei</i>	100	99	<i>sul2</i>	AAN06712.1
	722	<i>Shigella flexneri</i>	100	96	<i>sul2</i>	WP_025761012.1

Table 5 (continued)

Similarities with reported case						
Drug resistance gene	No. of nucleotides	Bacteria	Percentage similarity	Query Cover	Reported gene	Gene bank Accession No
	722	<i>Shigella flexneri</i>	100	93	<i>sul2</i>	EAA2119355.1
	699	<i>Shigella flexneri</i>	100	93	<i>sul2</i>	EAA2119355.1
	722	<i>Shigella flexneri</i>	100	93	<i>sul2</i>	WP_025761012.1
	722	<i>Shigella flexneri</i>	100	92	<i>sul2</i>	EGK20737.1
	711	<i>Shigella flexneri</i>	100	92	<i>sul2</i>	WP_025761012.1
	722	<i>Shigella flexneri</i>	100	92	<i>sul2</i>	EGK20737.1
	720	<i>Shigella sonnei</i>	100	91	<i>sul2</i>	AAN06712.1
	722	<i>Shigella sonnei</i>	100	91	<i>sul2</i>	WP_077125651.1
	722	<i>Shigella sonnei</i>	100	90	<i>sul2</i>	WP_077125651.1
	722	<i>Shigella sonnei</i>	100	89	<i>Sul2</i>	WP_077125651.1
	722	<i>Shigella flexneri</i>	100	86	<i>sul2</i>	WP_025761012.1
	722	<i>Shigella flexneri</i>	99.49	89	<i>sul2</i>	EGK20737.1

Table 6 Frequency of antibiotic resistance in *S. flexneri* isolated from patients using antibiotics according to prescription and without prescription

Category of patients	Positive, n (%)	Negative, n (%)	P value
Patients taken antibiotics according to prescription	55 (35.4)	100 (64.6)	< 0.0001
Patients taken antibiotics without prescription	100 (64.6)	55 (35.4)	

P value < 0.05 was considered as statistically significant

Table 7 Distribution of phenotypic antibiotic resistance profiles among patients using third-generation cephalosporin according to prescription and without prescription

Antimicrobial drugs	Total samples, N	With prescription, n (%)	Without prescription, n (%)	P value
CAZ (30 µg)	45	10 (22.3)	35 (77.7)	0.02
CPD (10 µg)	49	20 (40.8)	29 (59.2)	0.41
CFM (5 µg)	48	15 (31.25)	33 (68.8)	0.42
CRO (30 µg)	47	23 (48.93)	24 (51.1)	0.03

P value < 0.05 was considered statistically significant

CAZ Ceftazidime, CPD Cefpodoxime, CFM Cefixime, CRO Ceftriaxone

antibiotic drugs and that 95% of the *S. flexneri* isolates revealed some kind of MDR. Moreover, these multidrug-resistant isolates were concurrently resistant to three or more tested drugs out of the thirteen antimicrobial drugs used. Resistance to AMC,SXT,C (34%) was the most common pattern and resistance to at least 4, 5, 6, 7, 9

antimicrobial agents was also observed. The pattern of antimicrobial resistance among different serotypes of *S. flexneri* were investigated in our previous study [18]. The high level of multidrug-resistant frequency could be explained by overuse and misuse of the treatment of shigellosis using antimicrobial drugs over a long time which would lead to positive selection pressure that further limit the choice of antimicrobial drug in the treatment of shigellosis [10, 39].

Our data further explained the genetic background and mechanisms underlying *S. flexneri* resistance to chloramphenicol, third-generation cephalosporin, sulfa drug, and quinolones. Beta-lactamase genes (*bla*TEM and *bla*OXA) in *S. flexneri* were also detected by PCR and were subsequently sequenced. All *bla*TEM and *bla*OXA genes should produce TEM-1 and OXA-1 type beta-lactamases since no mutations in these genes were observed. Previously it was shown that the genes are plasmid-mediated [40, 41] which promotes beta-lactamase antibiotic-resistant by horizontal transfer [42, 43]. In the current study, all *bla*OXA and *bla*TEM isolates carried *bla*OXA-1 and *bla*TEM-1 suggesting specificity of host for *Shigella*, which is in parallel with previous studies [44, 45]. Among the studied genes the *bla*TEM gene is the only drug-resistant gene that was not previously sequenced in *S. flexneri* and showed close similarity with earlier described cases that were characterized in strains of *Escherichia coli*, *Klebsiella pneumoniae*, *Shigella sonnei*, *Acinetobacter baumannii*, and *Salmonella enterica* subsp.

Our results also revealed that the quinolone-resistant gene *qnrS*, encoded on a plasmid, was detected in all isolates while no mutation was identified. This indicates that the increased resistance to quinolone possibly may be due to the QRDR mutation which needs to be further investigated. In the current study, the sulfamethoxazole-resistant

gene, *sul2* was noticed in *S. flexneri* isolates and again no mutations were detected. None of *S. flexneri* sulfa-resistant isolates harbor *sul1* which indicates that in this study the resistant gene for sulfamethoxazole is associated with *sul2* gene which is inconsistent with the study reported in 2014 in Bangladesh [39]. In some *S. flexneri* isolates, drug-resistant genes were identified but were not resistant to that drug. This may be due to a poorer expression level of the precise gene in these specific isolates.

Finally, we observed a statistically significant association between patients who takes medication/antibiotic without prescription and patient who uses medication with prescription indicating that the indiscriminate use increases the resistance against third-generation cephalosporin and other antibiotics. Reduced resistance toward various drugs, by reducing the selection pressure on *S. flexneri* could, therefore, be accomplished if antibiotic use is better controlled.

The findings of this study are important because it showed the association of drug-resistant *S. flexneri* isolated from clinical and non-clinical sources in diarrhea endemic area of Pakistan, the Peshawar because there is no such data available in other areas of the country. This study had a limitation that we have studied selected drug resistance genes however; many other additional targets such as CTX-M, CMY, or QRDR (quinolone resistance determining regions) could have been included in the study.

Conclusion

In conclusion, increase burden of third-generation cephalosporin-resistant *S. flexneri* and also multidrug-resistant *S. flexneri* was observed in pediatrics which possesses a potential threat to the community in various areas of Peshawar, Pakistan. This increasing resistance against third-generation cephalosporin indicates that the choice of treatment has become challenging in the case of dysentery primarily in healthcare settings, where the public has already compromised health problems. This study suggested imipenem, piperacillin-tazobactam, and amikacin is a choice of treatment for shigellosis. Our data demonstrate the need to design a long-term surveillance program that would be essential in sorting out the alterations in antibiotic susceptibility profiles and the giving out of this data to clinicians. Also, it would be essential for the safe and effective use of antibiotics, to protect public health in this way. Clinicians should advice for antibiotic susceptibility testing before prescribing antibiotic treatment for patients. Government agencies should keep a check and balance on antibiotic supply and restrict antibiotic use by making policies and also by giving training to the health care workers applying antibiotic drug selection.

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Data Availability Not Applicable.

Code availability Not Applicable.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

Ethical Approval All procedures followed were in accordance with the ethical standards of the responsible committee on human experimentation (institutional and national) and with the Helsinki Declaration of 1975, as revised in 2000 approved by the Kohat University of Science and Technology, Kohat research and ethical committee.

Consent to Participate Informed consent was obtained from all patients/guardian for being included in the study.

Consent for Publication Not Applicable.

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